

IT4020 : Modern Topics in IT

Year 4 Semester 1/2

2026

Assignment Title	Assignment 1
Learning outcomes covered	LO1 – LO3
Assignment Mode	Take home Assignment
Maximum Marks	100
Contribution to the Final Grade	20%
Date published	09.02.2026
Deadline for submissions	20.02.2026
Mode of Submission	Report

Assignment Description

This is an individual assignment. This assignment focuses on experimenting with multiple GenAI tools, applying them to a practical IT-related problem and evaluating their performance.

Assignment Scenario

Assume you have been hired as a Junior AI Technologist in a software development company. The team wants to explore how generative AI can enhance:

- A. Software engineering tasks
- B. Data analysis and visualization
- C. UI/UX design

Task 1 – Practical Application

Choose one of the following categories and use two different Generative AI tools to complete the task. Include screenshots, prompts, generated outputs, and your own code (if applicable).

Option	Task	Deliverables
Option A – Software Engineering	• Generate code for a small functional module (e.g., login system, API handler, chatbot, CRUD app) • Debug an existing code snippet (you may choose any language) • Generate documentation for the module	• Generated code • Debugged output • Documentation created by the AI • Your manual improvements
Option B – Data Analytics & Visualization	• Clean or transform a dataset (any dataset of your choice) • Generate visualizations • Produce an analytical summary or insights report	• Prompts used • Generated scripts/visualizations • Your validation of correctness
Option C – UI/UX & Design	• A UI wireframe • A graphical mock up • A user persona and scenario	• Tool outputs • Your redesign or enhancements

Task 2 – Comparative Evaluation

Compare the two GenAI tools you used based on:

- Output Quality
- Prompt Sensitivity
- Technical Limitations
- Performance Metrics

Marking Scheme

Criteria	Excellent (90-100)	Good (70-80)	Satisfactory (50-60)	Poor (30-40)	Very Poor (0-20)
Quality of Prompts (20 marks)	Demonstrates excellent prompt engineering; prompts are clear, specific, optimized, display constraints, examples, and role definitions. Shows iterative refinement.	Prompts are good, well-structured, with attempts at refinement and constraint use.	Basic prompts used; limited refinement; missing advanced structures.	Vague or generic prompts; little evidence of experimentation.	Minimal prompts or incorrect use; lacks understanding of prompting.
Quality of Generated Outputs (20 marks)	Outputs are highly relevant, functional, accurate; student validates correctness; improvements applied.	Mostly correct outputs; small issues corrected by student.	Mixed quality outputs; some inaccuracies not identified.	Outputs contain errors; student fails to address them.	Outputs incorrect and unexamined.
Technical Implementation & Validation (20 marks)	Thorough validation with manual checking, testing, debugging, dataset checks, code execution, etc. Evidence of deep technical understanding.	Good validation with explanations; minor gaps.	Some validation attempted; lacks depth.	Minimal validation; poor understanding of errors.	No validation or technical explanation.
Output Quality Comparison (10 marks)	Insightful, well-structured comparison with concrete examples; deep analysis.	Good comparison with clear examples.	Basic comparison; lacks depth.	Weak analysis with minimal evidence.	No meaningful comparison provided.

Department of Information Technology
Faculty of Computing
Sri Lanka Institute of Information Technology



Sensitivity to Prompts (10 marks)	Extensive testing and clear demonstration of prompt sensitivity; excellent interpretation.	Good variety of prompt modifications; adequate analysis.	Some prompt variations tested; lacks depth.	Minimal variations; weak discussion.	No meaningful tests of prompt sensitivity.
Comparison of GenAI tools (10 marks)	Provides a thorough, well-structured comparison of tools using clear criteria (accuracy, usability, limitations, ethics, multimodal capabilities, speed, reasoning, cost).	Compares tools with generally clear criteria and examples. Shows reasonable understanding of differences.	Basic comparison between tools; may lack consistency in evaluation criteria or depth. Limited examples.	Weak or superficial comparison; criteria unclear, inconsistent, or missing; little supporting evidence.	No meaningful comparison; descriptions incorrect, incomplete, or missing.
Viva (10 marks)	Demonstrates clear understanding of GenAI tools, outputs, prompts, methodology, and comparison analysis. Answers confidently, explains reasoning, justifies choices, and reflects on limitations. Able to respond to follow-up questions with depth and accuracy.	Shows good understanding; explains work reasonably well; answers most questions; minor misconceptions.	Understands the basics but struggles to justify decisions or explain technical reasoning; answers show gaps.	Weak understanding; cannot explain key parts of their own work; inconsistent responses; relies on memorization.	Very poor understanding; cannot explain methodology, prompts, outputs, decisions, or comparisons; indicates limited or no authorship of submitted work.